IE4728 Enterprise Network Design

Enterprise Network Design Project

Design Project Implementation
(Week 8-10 / 11-13)

Student Groups: F52, F53

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Design Implementation
&
Simulation

Design Requirements (recap)

- **Network Requirements**
  - Interworking of workgroups
  - Network resource sharing (file/data, servers, printers)
- **User Requirements**
  - 4 Workgroup: Academic (Acad), Administrative (Admin), Research students (RS), and lab users (Lab)
  - Users collaborate within same workgroup and across different network segments
  - Lab groups do not interoperate nor share resources with other groups
- **Network Design features:**
  - **LAN segmentation** (and augmentation) of physical networks
  - **VLSM subnetting** (and supernetting) of logical networks
  - **Optimized** layered design for performance and robustness
- **Access control lists for:**
  - Host and network access
  - Interworking of workgroups
  - Internetworking and resource sharing

Design Work (recap)

- **Design approach**
  - Top-down analysis of design requirements
  - Bottom-up implementation of layered network design
  - Five (5) design templates to guide design and implementation
- **Network design model**
  - Modular and hierarchical layered design: core, distribution, and access layers
- **Design implementation**
  - Minimize address wastage while maximizing route aggregation.
  - Optimized load distribution and robustness against node/link failures
- **Access control policies**
  - ACLs for internetworking of subnets and interworking of workgroups
- **Network design simulations**
  - Simulation of design and implementation using Packet Tracer

Design Project Assessment Rubric

No	Assessment Items	%
1	Top-Down analysis of requirement (subnetting plan)	20
2	Bottom-Up implementation of layered network (supernetting)	10
3	Internetworking of LANs and WAN (routing)	10
4	Access Control Lists for internetworking and interworking	10
5	Network simulation using Packet Tracer v8	
	• IP address assignment	10
	• Network routing	10
	• Access Control	10
6	Report and Slides	15
7	Weekly progress	5
	Total percentage	100

Project Deliverables and Submission (recap)

1. **Design Project report**
   - One softcopy of the report to be submitted to Turnitin
2. **Packet Tracer simulation***
   - One simulation file generated using Packet Tracer v8
3. **PowerPoint slides***
   - one power-point file summarizes the network design

   * submit both files as one assignment

- **Submit as Assignment on course site**
  - Name reports/files in the format, “Fxx-Gyy_type”
  - Fxx is tutorial group, Gyy is design group, and type is “report”, “simulation”, or “slides” as appropriate, e.g. F51-G02_Report
- **Submission deadline**
  - one week after the 3-week design class (i.e. the 4th week)

Project Completion Schedule

Reasonable weekly progress:

- Week 1: Completion of Table 1
  - Top-down analysis of network requirements
  - Completed LAN segmentation of physical networks
  - Completed subnetting and supernetting of logical networks
- Week 2: Completion of Table 2 and 3
  - Bottom-up network implementation of logical networks
  - Completed Host IP assignment and server placement
  - Completed inter-router address assignment
- Week 3: Completion of Table 4 and 5
  - Access control lists (ACL) for network security and data sharing
  - Completed design and placement of ACLs
  - Created simulation scenarios for testing the network design

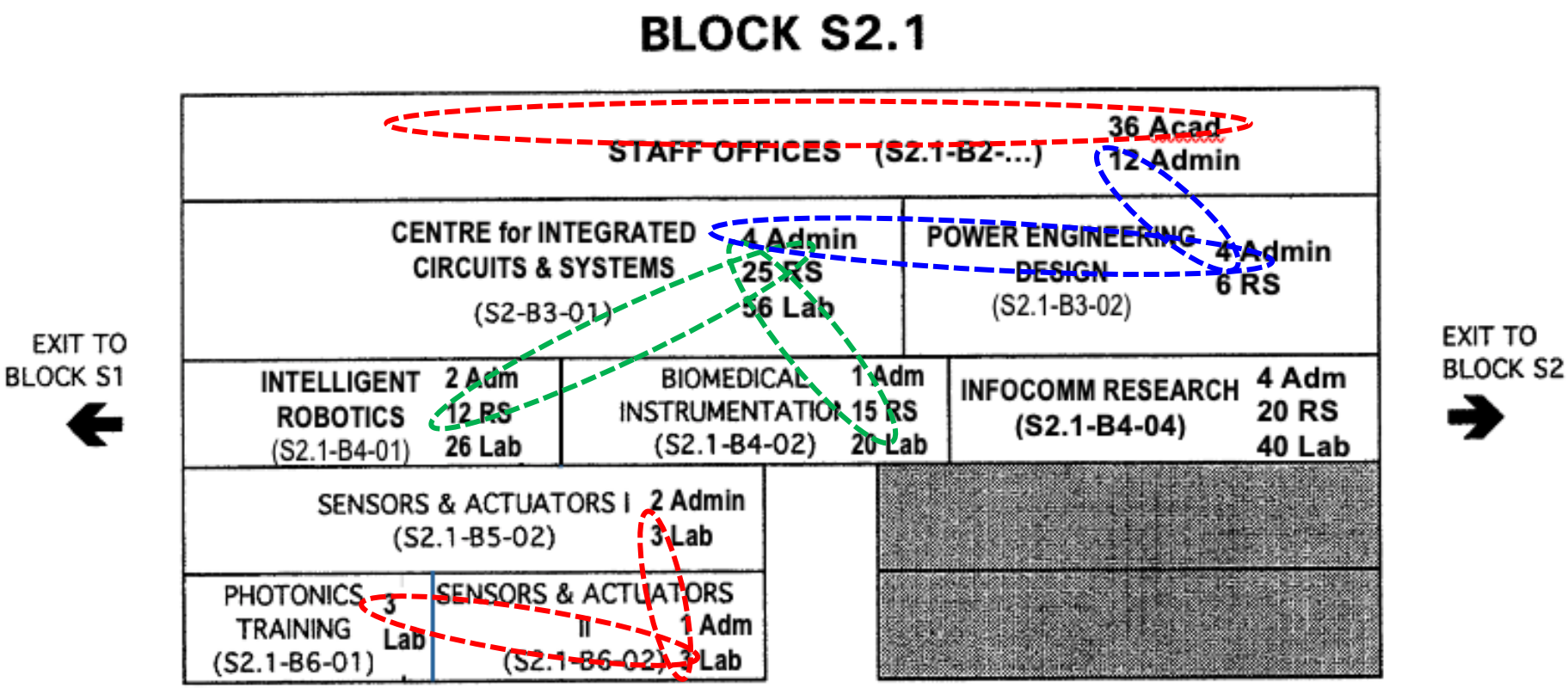
Week 1

LAN Design

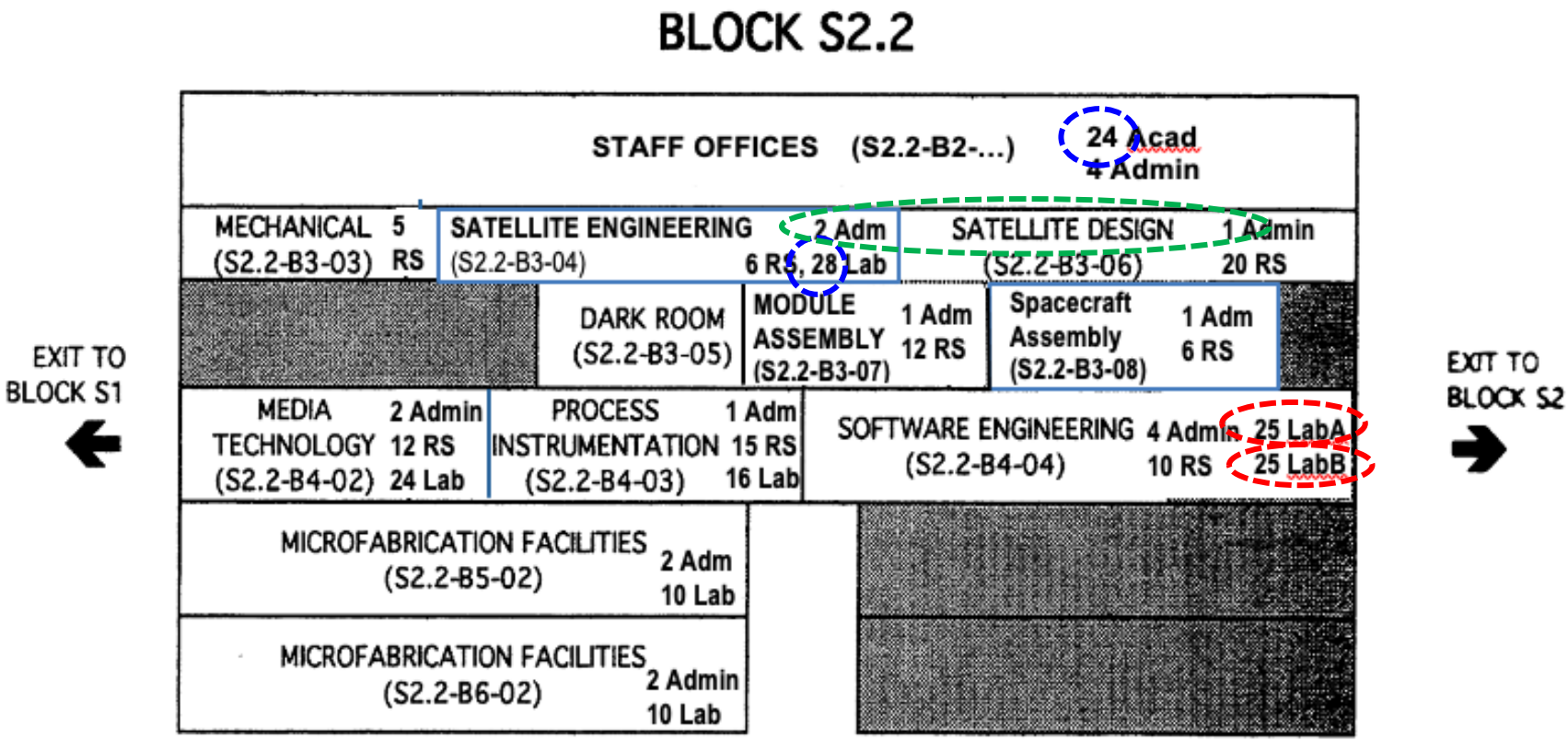
LAN Segmentation and Augmentation

- LAN Segmentation
  - Partition a LAN into smaller LAN segments to conserve address space
  - Split a LAN segment into smaller segments, wherever more addresses can be conserved than otherwise wasted
- LAN Augmentation
  - Merge LAN segments into a larger LAN to minimize address wastage
  - Form a larger LAN from among smaller LAN segments of the same workgroup, unless constrained by maximum network diameter
- Maximum Network Diameter
  - Maximum segment length spanning no more than 2 adjoining rooms on the same floor or on the same stack

Distribution of Hosts and User Groups in S2.1



Distribution of Hosts and User Groups in S2.2



Shared Printers

- Shared Printer
  - Each user group in a room shares a printer among its own group
  - A printer is shared either as a local printer or a network printer.
- Local vs Network Printer
  - A local printer is shared within its LAN segment.
  - A network printer is shared across networks on a WAN
- Number of Printers
  - Each user group in a room shares a printer
    - Same user groups do not share a printer across rooms
  - A user group in a larger room shares more than 1 printer
    - For convenience of distance, a room larger than 2-room length requires more printers

Shared Servers

- Data vs File Servers
  - Data are not shared across different labs
    - Local server shares data among users connected to a LAN
  - Files may be shared across different workgroups
    - Network server shares files among users connected across different LANs
- Group Server
  - A file server shared within a workgroup.
- Enterprise Server
  - A server shared across all workgroups in an enterprise network.
- Server Farm
  - A group of servers connected to a LAN segment.
- Server Placement
  - A server is placed at the least equal network distance to all its users, and at a network layer that it does not cause traffic flooding into other layers

Table 1a: Template for Network Address Assignments

Template for Network Address Assignments using VLSM Subnetting								
Location	Acad		Admin		RS		Lab	
Block S2.1	Hosts	Net#/Subnet Bits	Hosts	Net#/Subnet Bits	Hosts	Net#/Subnet Bits	Hosts	Net#/Subnet Bits
S2.1-b2	36	x.x.28.01/26	12					
s2.1-b3-01			4		25		56	x.x.30.01/26
s2.1-b3-02			4		6			
s2.1-b4-01			2		12		26	
s2.1-b4-02			1		15		20	
s2.1-b4-04			4		20		40	
s2.1-b5-02			2				3	x.x.30.0010/28
s2.1-b6-01							3	
s2.1-b6-02			1				3	x.x.30.0010/28
Total hosts*	**36+n**		**30**		**78**		**151**	
Block S2.2								
sS.2-b2	24	x.x.28.001/27	4					
s2.2-b3-03					5			
s2.2-b3-04			2		6		28	
s2.2-b3-06			1		20			
s2.2-b3-05								
s2.2-b3-07			1		12			
s2.2-b3-08			1		6			
s2.2-b4-02			2		12		24	
s2.2-b4-03			1		15		16	
s2.2-b4-04			4		10		2x25	
s2.2-b5-02			2				10	
s2.2-b6-02			2				10	
Total hosts*	**24+n**		**20**		**86**		**138**	
* to include additional n addresses for printers, servers, and router interfaces.								

All values are for illustration only

Table 1b: Template for Network Address Summarization

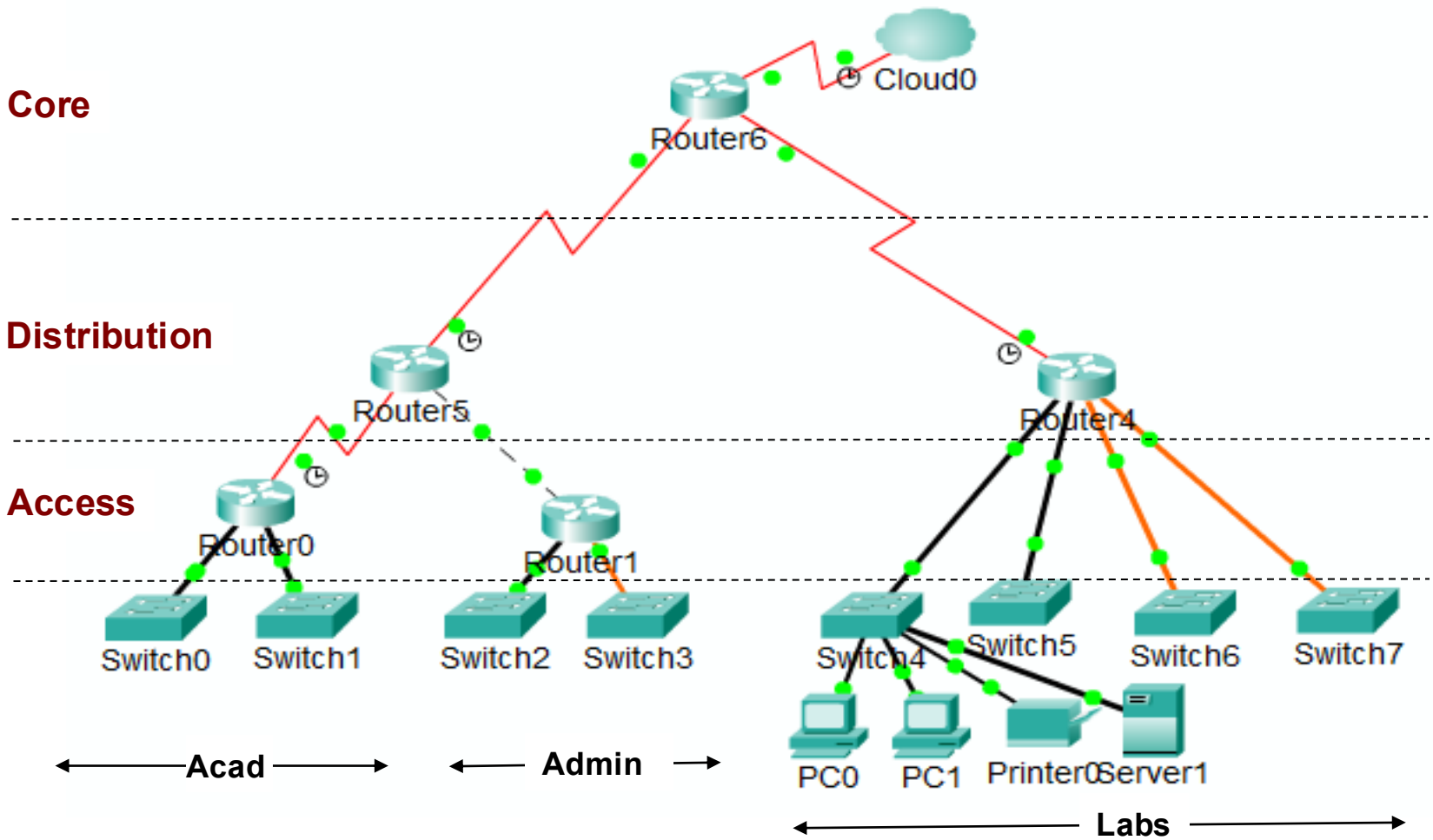
Supernetting of Subnets for Network Address Summarization			
Network	Net#/Supernet Bits	Subnets	Description of logical group assignments
x.x.28.0	x.x.28.0/25	x.x.28.01/26 x.x.28.001/27	Supernet for aggregration of Acad groups in S2.1 and S2.2
			<Admin groups>
			<RS groups in S2.1>
			<RS groups in S2.2>
x.x.30.0			Network for all Lab groups in S2.1
x.x.30.32/28			Subnet for combined Lab groups in S2.1-b5-02 & S2.1-b6-02
			<All Lab groups in S2.2>
			<Combined Lab groups in S2.2>

Table 1b. Sample template for network address summarization

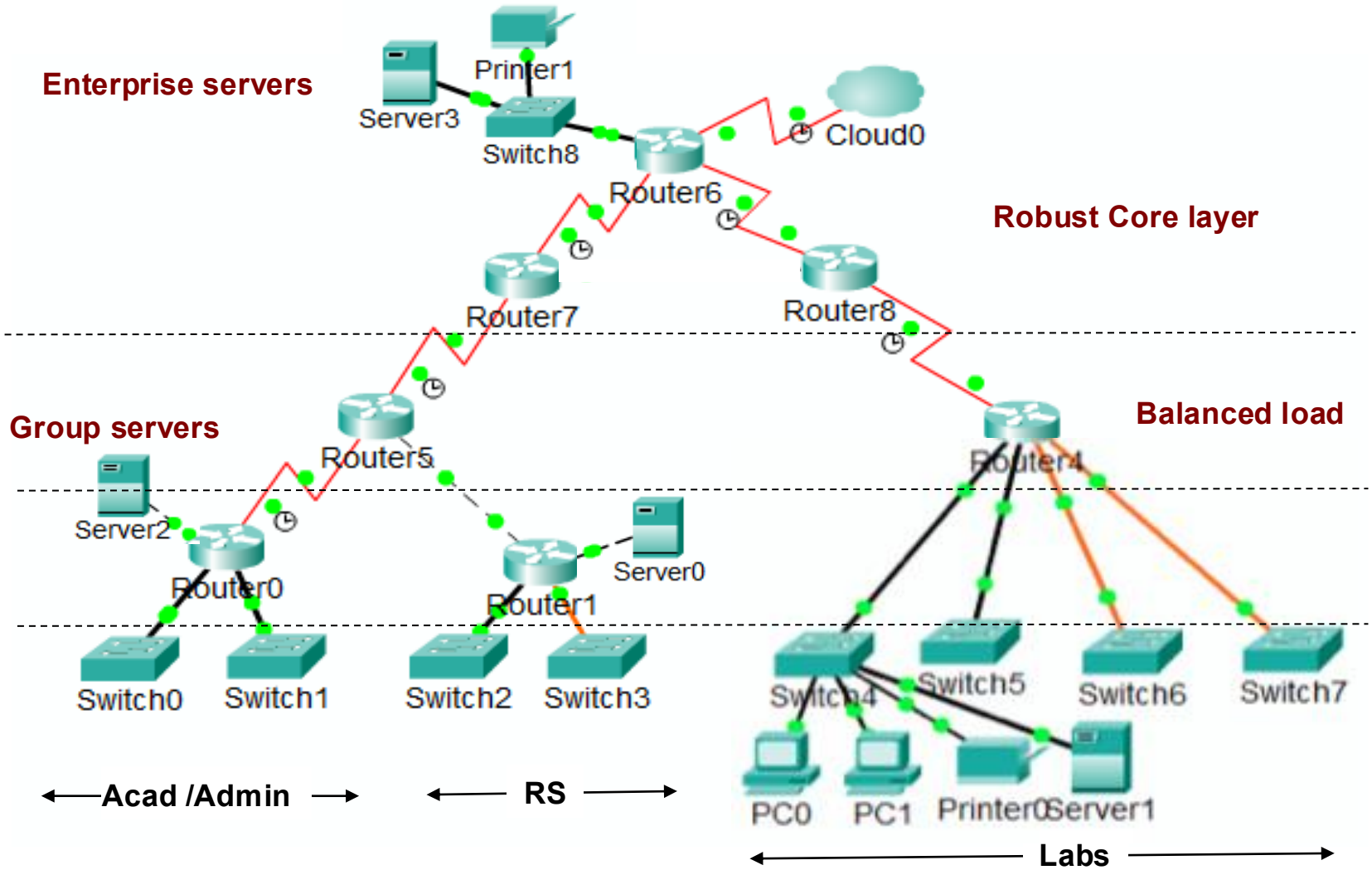
Week 2

WAN Design

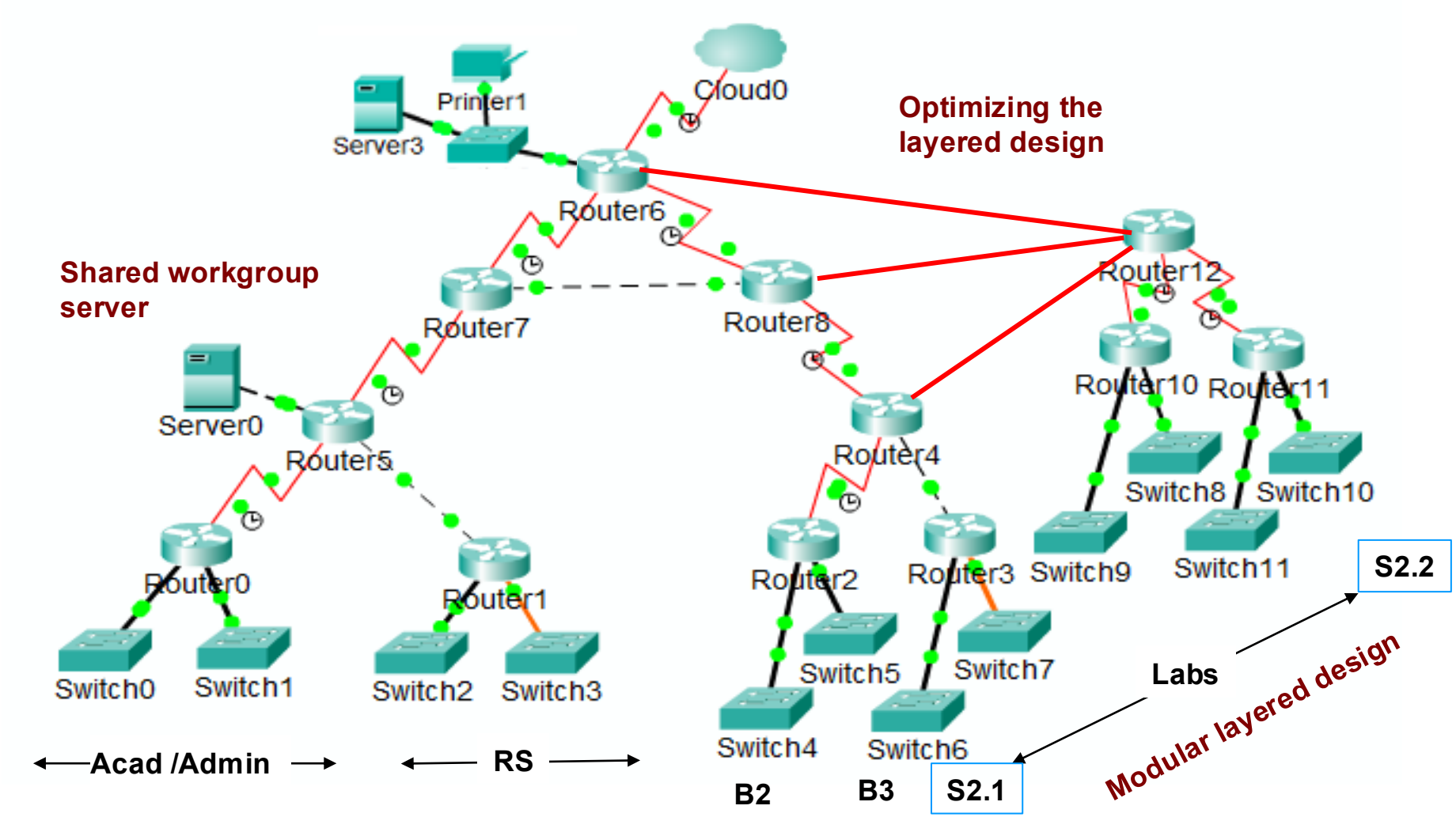
A 3-layered Enterprise Network



Load Distribution & Server Placement



Modular & Layered Design



Node & Link Redundancies

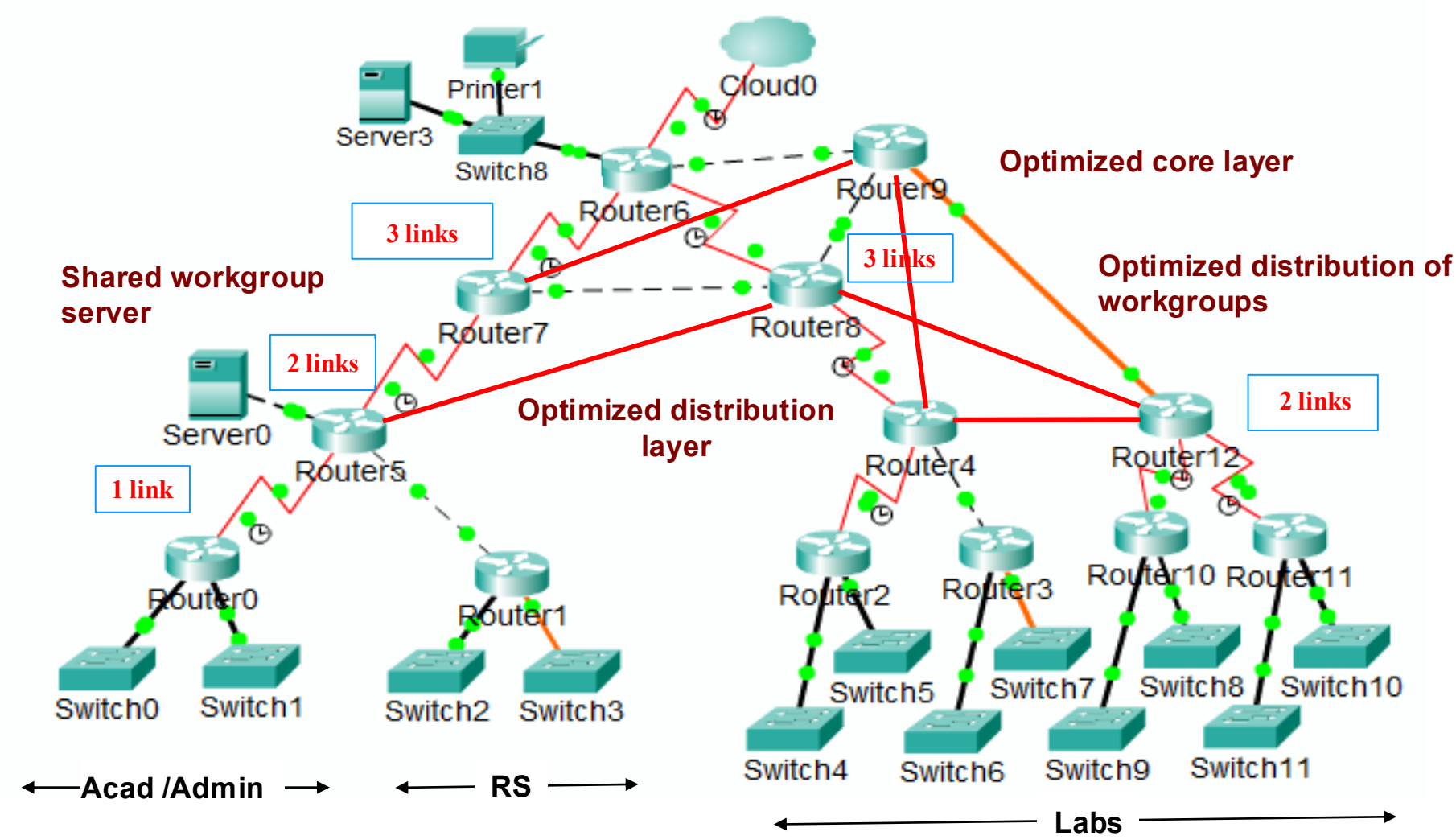


Table 2: Template for Host IP Address Assignments

User Group	Locations	No of Hosts	No. of Printer/ Server	Sub Network Address	Sub-net Mask	Router Address	First Host Address	Last Host Address	Printer Address	Server Address	No. used / unused
Lab_2.1	CICS	56	1 / 1	207.21.30.0	/26	x.x.x.65	x.x.x 66	x.y.z 121	x.x.x.122	x.x.x.123	59 / 3
	Intelligent Robots	26									
	Biomedical	20									
	Infocommunication Research	40									
	Sensors & Actuators i	3	1/1		2x /29 or /28 ?						2x 6/0 or 12/2 ?
	Sensors & Actuators II	3	1/0								
	Photonics	3									

Table 2. Table template for host address assignments

- This template records local devices only
  - printers and servers are directly connected to the respective LAN segments via switches.

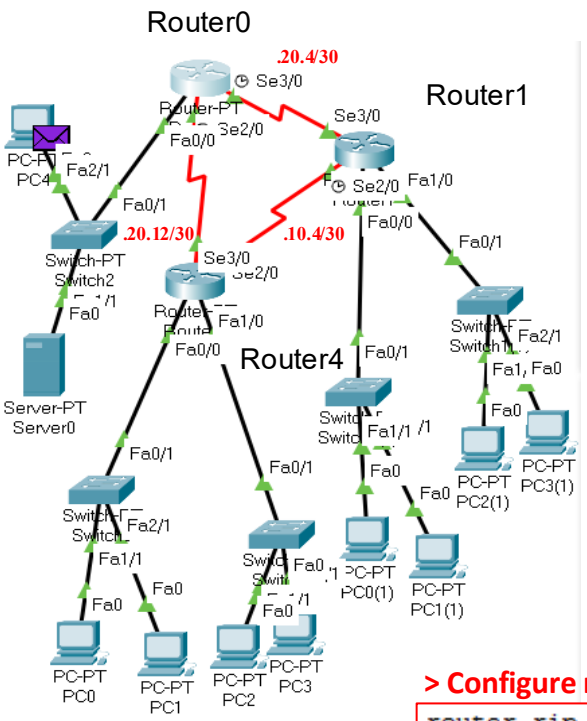
Tables 3: Templates for Server Address Assignment and Placement

Server Name	Server IP Address	Subnet Mask	Attached Router Name	Router Interface
HTTP server				
DNS server				
SQL Server				
Acad Server				
Admin Server				
RS Server				
Email Server				
(all ntwk devices)				
Etc…				

Table3. Sample template for server address assignments

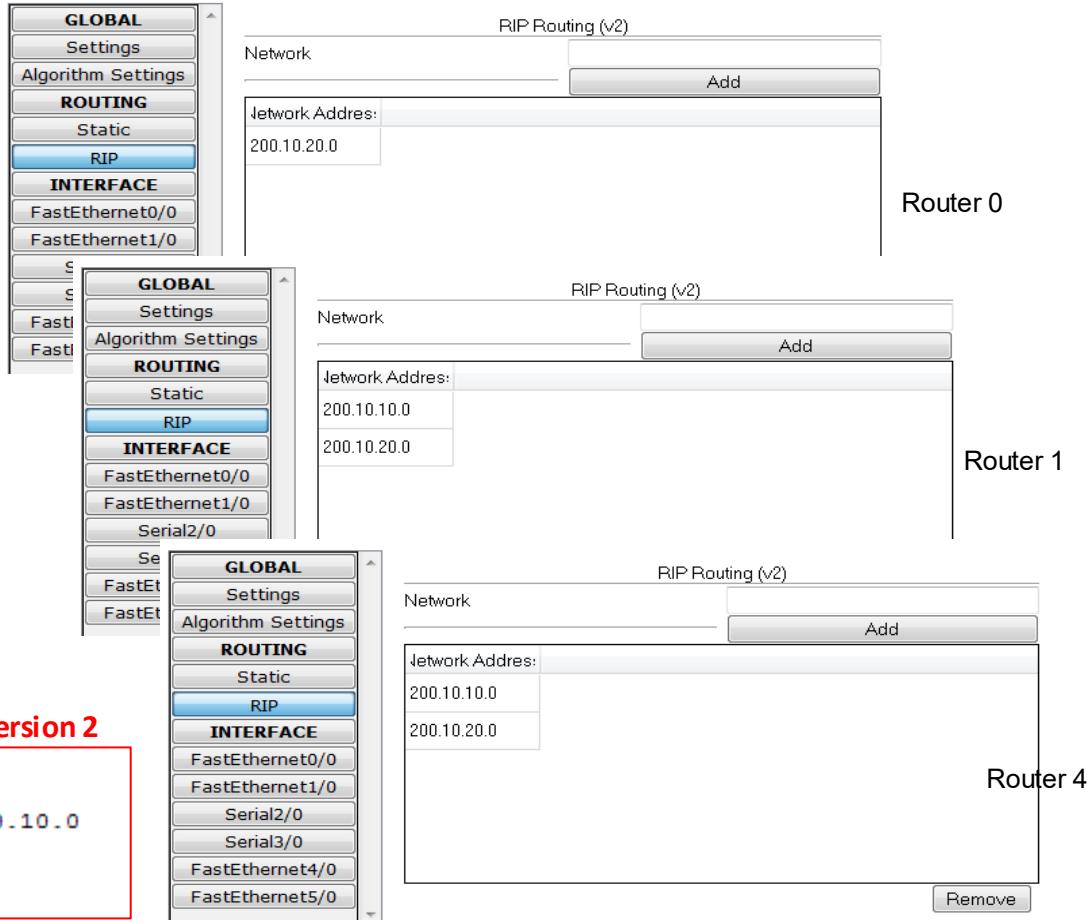
- This template records network devices only
  - network printers and servers are connected to subnets via routers
- A server may host multiple services identified by port numbers
- Server at different layers of a network
  - Access layer - local servers
  - Distribution layer - group servers
  - Core layer – enterprise servers / server farm

Configuring the WAN using RIPv2

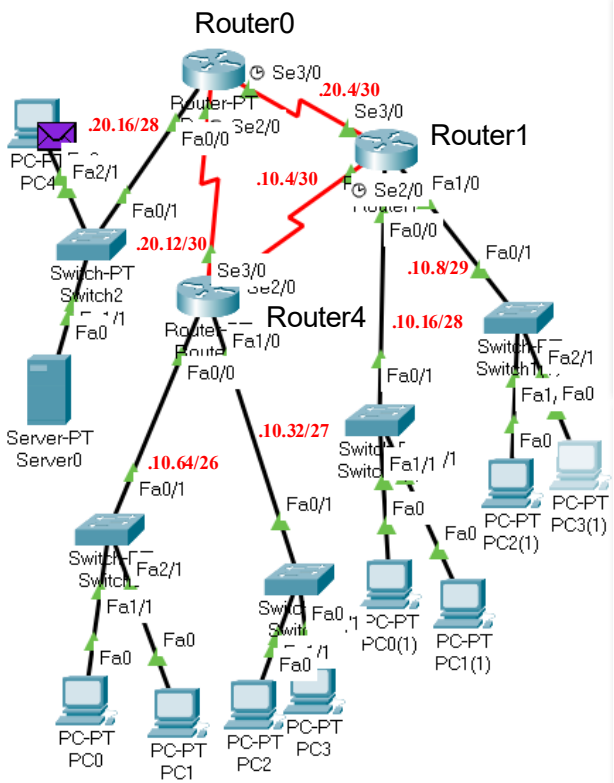


> Configure rip version 2

```
router rip
 version 2
 network 200.10.10.0
!
ip classless
!
```



Auto-Summary of Routes



Routing Table for Router 0

Type	Network	Port	Next Hop IP	Metric
R	200.10.10.0/24	Serial3/0	200.10.20.6	120/4
R	200.10.20.0/24	Serial2/0	200.10.20.14	120/2
C	200.10.20.4/30	Serial3/0	---	0/0
C	200.10.20.12/30	Serial2/0	---	0/0
C	200.10.20.16/28	FastEthernet0/0	---	0/0

Routing Table for Router4

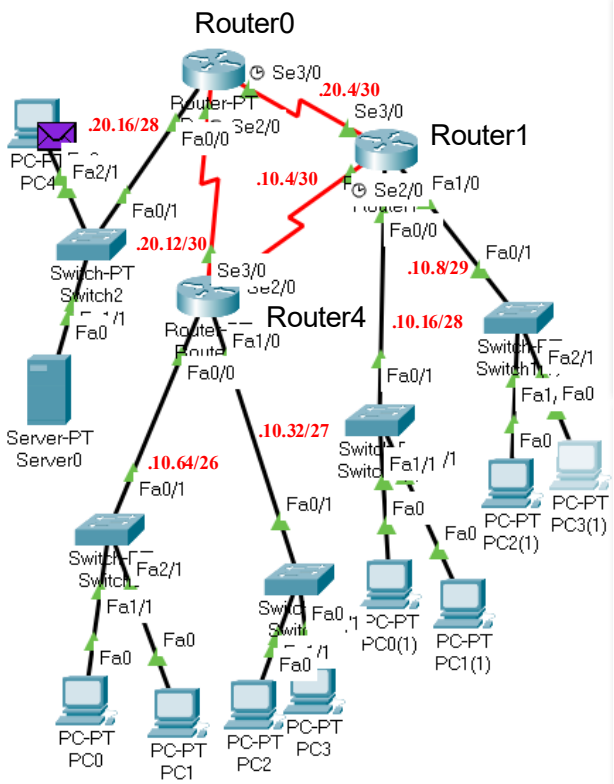
Type	Network	Port	Next Hop IP	Metric
R	200.10.10.0/24	Serial2/0	200.10.10.6	120/6
C	200.10.10.4/30	Serial2/0	---	0/0
R	200.10.10.8/29	Serial2/0	200.10.10.6	120/1
R	200.10.10.16/28	Serial2/0	200.10.10.6	120/1
C	200.10.10.32/27	FastEthernet1/0	---	0/0
C	200.10.10.64/26	FastEthernet0/0	---	0/0

Routing Table for Router 1

Type	Network	Port	Next Hop IP	Metric
R	200.10.10.0/24	Serial3/0	200.10.20.5	120/11
C	200.10.10.4/30	Serial2/0	---	
C	200.10.10.8/29	FastEthernet1/0	---	0/0
C	200.10.10.16/28	FastEthernet0/0	---	0/0
R	200.10.10.32/27	Serial2/0	200.10.10.5	120/1
R	200.10.10.64/26	Serial2/0	200.10.10.5	120/1
R	200.10.20.0/24	Serial2/0	200.10.10.5	120/1
C	200.10.20.4/30	Serial3/0	---	0/0
R	200.10.20.12/30	Serial3/0	200.10.20.5	120/1
R	200.10.20.16/28	Serial3/0	200.10.20.5	120/1

Screensho

No Auto-Summary of Routes



Routing Table for Router0

Type	Network	Port	Next Hop IP	Metric
R	200.10.10.4/30	Serial2/0	200.10.20.14	120/1
R	200.10.10.4/30	Serial3/0	200.10.20.6	120/1
R	200.10.10.8/29	Serial3/0	200.10.20.6	120/1
R	200.10.10.16/28	Serial3/0	200.10.20.6	120/1
R	200.10.10.32/27	Serial2/0	200.10.20.14	120/1
R	200.10.10.64/26	Serial2/0	200.10.20.14	120/1
C	200.10.20.4/30	Serial3/0	—	0/0
C	200.10.20.12/30	Serial2/0	—	0/0
C	200.10.20.16/28	FastEthernet0/0	—	0/0

Routing Table for Router4

Type	Network	Port	Next Hop IP	Metric
C	200.10.10.4/30	Serial2/0	—	0/0
R	200.10.10.8/29	Serial2/0	200.10.10.6	120/1
R	200.10.10.16/28	Serial2/0	200.10.10.6	120/1
C	200.10.10.32/27	FastEthernet1/0	—	0/0
C	200.10.10.64/26	FastEthernet0/0	—	0/0
R	200.10.20.4/30	Serial3/0	200.10.20.13	120/1
R	200.10.20.4/30	Serial2/0	200.10.10.6	120/1
C	200.10.20.12/30	Serial3/0	—	0/0
R	200.10.20.16/28	Serial3/0	200.10.20.13	120/1

Routing Table for Router1

Type	Network	Port	Next Hop IP	Metric
C	200.10.10.4/30	Serial2/0	—	0/0
C	200.10.10.8/29	FastEthernet1/0	—	0/0
C	200.10.10.16/28	FastEthernet0/0	—	0/0
R	200.10.10.32/27	Serial2/0	200.10.10.5	120/1
R	200.10.10.64/26	Serial2/0	200.10.10.5	120/1
C	200.10.20.4/30	Serial3/0	—	0/0
R	200.10.20.12/30	Serial3/0	200.10.20.5	120/1
R	200.10.20.12/30	Serial2/0	200.10.10.5	120/1
R	200.10.20.16/28	Serial3/0	200.10.20.5	120/1

Week 3

ACL Design

Tables 4: Template for Assignment and Placement of Access Control Lists

Network Name	Router Name	Router Interface	Network #	Subnet mask	IP Address	Access List Number
RS Network	RS S2.1	0/0				x
		1/0				x
		2/0				1 (OUT)
	RS S2.2	0/0				x
		1/0				x
		2/0				x
	RS	0/0				x
		1/0				123 (IN)
		2/0				
		3/0				

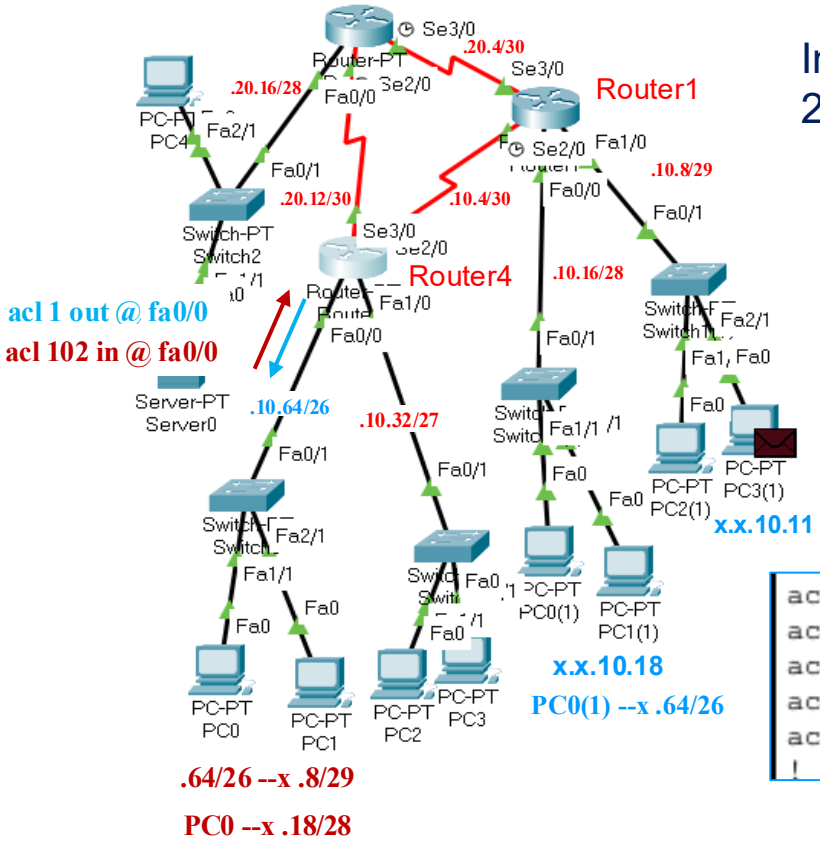
Table 5: Template for Access Control List Statements

Network	Router Name	Router Interface	IP Address	Type	Access-List
RS Network	RS S2.2	1/0		In/Out	Extended IP access list 101 (Out) **// permit FTP service from RS to Acad** permit tcp x.x.x.x filter y.y.y.y filter eq 21 **// deny FTP service from RS to others** deny tcp .. .. .. any eq 21 **// permit the rest access** permit ip any any
	RS	0/0		In/Out	Extended IP access list 102 (In) **// permit FTP service from Acad to RS** ... **// deny FTP service from others to RS** ... **// permit the rest access** ...

ACL – Standard deny source host and extended deny destination subnet at egress Router4

Standard ACL deny IP from a host 200.10.10.18/28 on Router1 to any host on subnet 200.10.10.64/26 on Router4.

In reverse direction, Router4 denies source subnet 200.10.10.64/26 to destination subnet 200.10.10.8/29

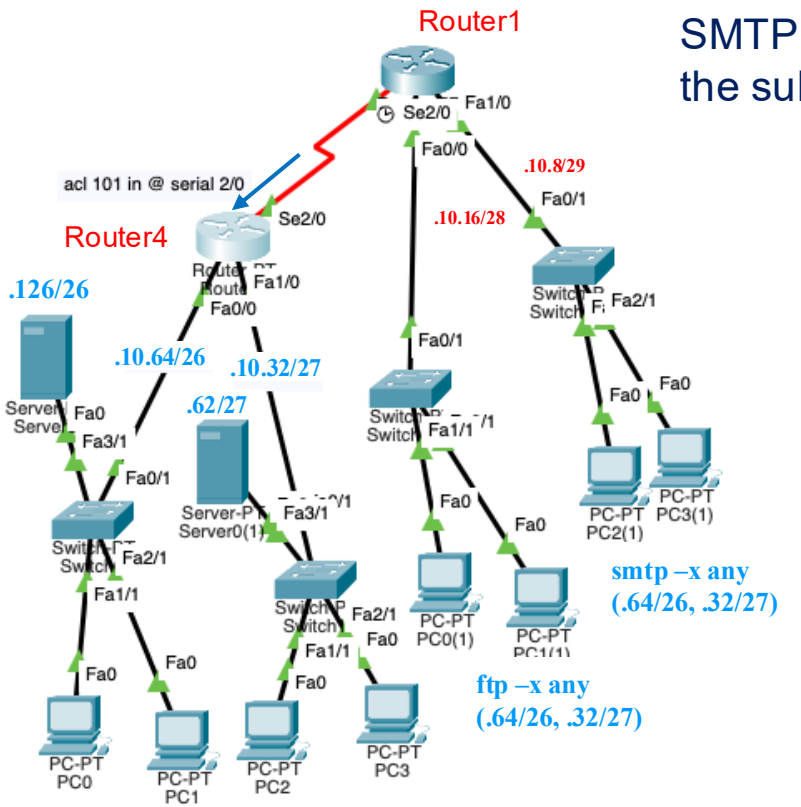


Router4

```
interface FastEthernet0/0
 ip address 200.10.10.65 255.255.255.192
 ip access-group 102 in
 ip access-group 1 out
 duplex auto
 speed auto
!
```

```
access-list 1 deny host 200.10.10.18
access-list 1 permit any
access-list 101 deny ip 200.10.10.64 0.0.0.63 200.10.10.8 0.0.0.7
access-list 102 deny ip 200.10.10.64 0.0.0.63 200.10.10.8 0.0.0.7
access-list 102 permit ip any any
!
```

ACL – Extended list to deny tcp destination subnets on egress Router4 (1)



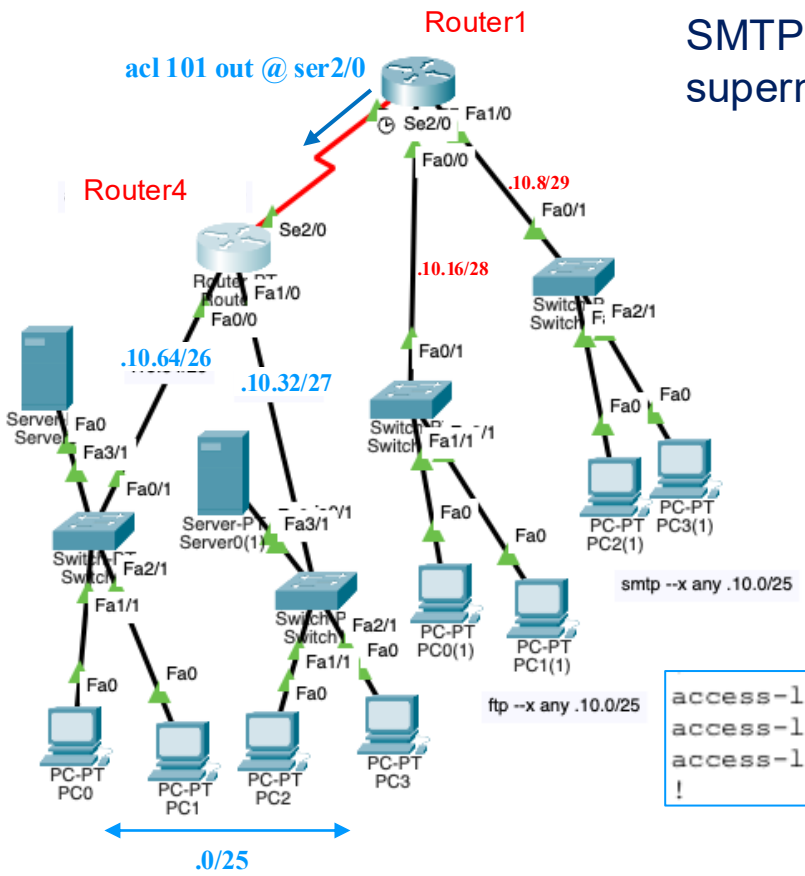
Extended ACL to deny FTP from subnet 200.10.10.16/28 and SMTP from subnet 200.10.10.8/29, both on Router1 to any of the subnets on Router4

Router4

```
interface Serial2/0
 ip address 200.10.10.5 255.255.255.252
 ip access-group 101 in
!
```

```
access-list 1 deny host 200.10.10.18
access-list 1 permit any
access-list 101 deny tcp 200.10.10.16 0.0.0.15 any eq ftp
access-list 101 deny tcp 200.10.10.8 0.0.0.7 any eq smtp
access-list 101 permit ip any any
!
```

ACL – Extended list to deny tcp destination supernet on ingress Router1 (2)



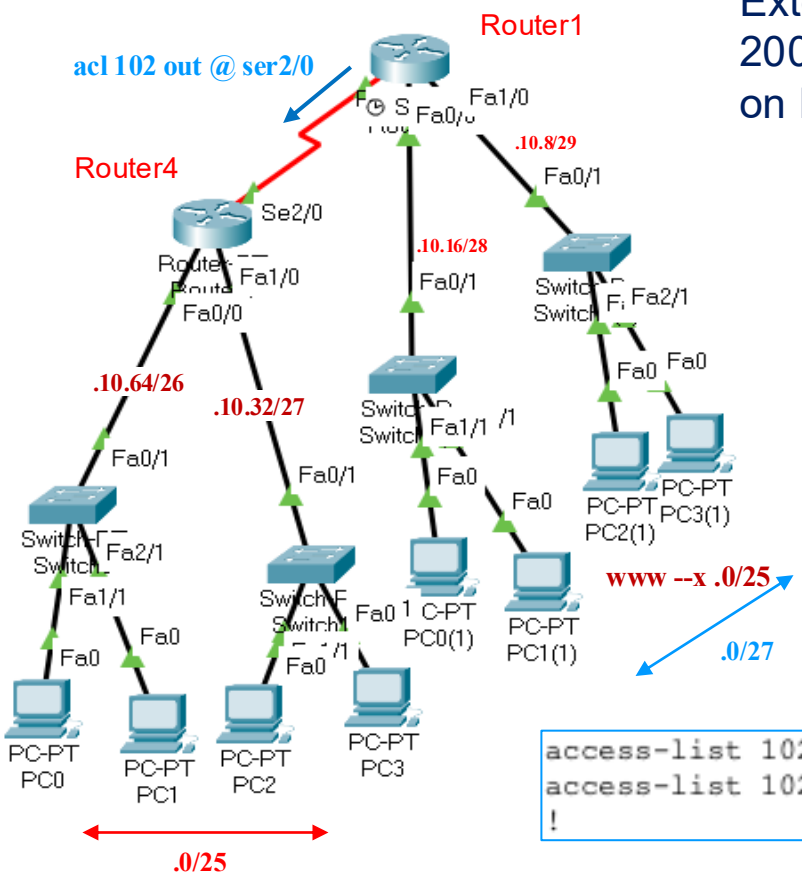
Extended ACL to deny FTP from subnet 200.10.10.16/28 and SMTP from subnet 200.10.10.8/29, both on Router1 to supernet 200.10.10.0/25 on Router4

Router1

```
interface Serial2/0
 ip address 200.10.10.6 255.255.255.252
 ip access-group 101 out
 clock rate 2000000
!
```

```
access-list 101 deny tcp 200.10.10.16 0.0.0.15 200.10.10.0 0.0.0.127 eq ftp
access-list 101 deny tcp 200.10.10.8 0.0.0.7 200.10.10.0 0.0.0.127 eq smtp
access-list 101 permit ip any any
!
```

ACL - Extended list to deny source supernet on ingress Router1 to destination supernet on Router4 (3)



Extended ACL to deny WWW service from supernet 200.10.10.0/27 on Router1 to supernet 200.10.10.0/25 on Router4

Router1

```
interface Serial2/0
 ip address 200.10.10.6 255.255.255.252
 ip access-group 102 out
 clock rate 2000000
!
```

```
access-list 102 deny tcp 200.10.10.0 0.0.0.31 200.10.10.0 0.0.0.127 eq www
access-list 102 permit ip any any
!
```